

INTERNAL AI AGENT EVALUATION REPORT

Executive Summary

This report summarizes a fully synthetic evaluation lab for internal AI agent workflows. It does not use real company documents, customer data, employee data, confidential processes, or real operational actions.

- Golden retrieval cases: 288
- Synthetic ticket extraction and agent cases: 180
- Red-team safety cases: 60
- Best current retriever: Hybrid sparse semantic retrieval
- Current vector experiments: local TF-IDF vector retrieval and local embedding-store retrieval

Retrieval Evaluation

System	Hit rate@3	Citation coverage	Next action accuracy	Abstention accuracy	Failures
Baseline team hints	45.54%	20.09%	20.09%	78.47%	241
Improved lexical	99.11%	98.66%	98.66%	100.00%	3
Hybrid sparse semantic	100.00%	100.00%	100.00%	100.00%	0
Local TF-IDF vector	100.00%	100.00%	100.00%	100.00%	0
Local embedding store	100.00%	100.00%	100.00%	100.00%	0

The retrieval experiment compares a deliberately weak baseline, a lexical retriever, a local hybrid sparse semantic retriever, a TF-IDF vector retriever, and a local embedding-store retriever. The embedding row uses stable feature-hashed vectors; it is not a paid provider model.

Retriever Metric Snapshots

Snapshot	System	Citation coverage	Failed cases	Citation delta	Failure delta	Regression	Reason
001_baseline_team_hints	Baseline team hints	20.09%	241			False	
002_improved_lexical	Improved lexical	98.66%	3	+78.57%	-238	False	
003_hybrid_sparse_semantic	Hybrid sparse semantic	100.00%	0	+1.34%	-3	False	
004_local_tf_idf_vector	Local TF-IDF vector	100.00%	0	+0.00%	+0	False	
005_local_embedding_store	Local embedding store	100.00%	0	+0.00%	+0	False	

Historical Evaluation Snapshots

Milestone time	Milestone	Citation coverage	Failed cases	Citation delta	Failure delta
2026-06-02T09:00:00Z	Baseline team hints	20.09%	241		
2026-06-02T10:00:00Z	Improved lexical	98.66%	3	+78.57%	-238

2026-06-02T11:00:00Z | Hybrid sparse semantic | 100.00% | 0 | +1.34% | -3
2026-06-02T12:00:00Z | Local TF-IDF vector | 100.00% | 0 | +0.00% | +0
2026-06-02T13:00:00Z | Local embedding store | 100.00% | 0 | +0.00% | +0

Retriever Failure Analysis

System | Failed cases | Retrieved but not cited | Abstention mismatches | Top failure reason

Baseline team hints | 241 | 57 | 62 | missing_or_wrong_citation (179)

Improved lexical | 3 | 1 | 0 | missing_or_wrong_citation (3)

Hybrid sparse semantic | 0 | 0 | 0 |

Local TF-IDF vector | 0 | 0 | 0 |

Local embedding store | 0 | 0 | 0 |

System | Case | Noise | Failure | Expected citation | Predicted citation | Retrieved but not cited | Top retrieved scores | Recommended fix

Baseline team hints | GOLD-TCK-0001 | clean_exact | missing_or_wrong_citation, wrong_issue_category, wrong_next_action | RB-TRADE_SUPPORT-02 | RB-TRADE_SUPPORT-01 | True | RB-TRADE_SUPPORT-01 total=3; RB-TRADE_SUPPORT-02 total=3; RB-TRADE_SUPPORT-03 total=3 | Add within-team reranking using issue-category evidence and expected action terms.

Baseline team hints | GOLD-TCK-0003 | clean_exact | missing_or_wrong_citation, wrong_issue_category, wrong_next_action | RB-PAYMENTS_OPS-03 | RB-PAYMENTS_OPS-01 | True | RB-PAYMENTS_OPS-01 total=4; RB-PAYMENTS_OPS-02 total=4; RB-PAYMENTS_OPS-03 total=4 | Add within-team reranking using issue-category evidence and expected action terms.

Baseline team hints | GOLD-TCK-0004 | clean_exact | missing_or_wrong_citation, wrong_issue_category, wrong_next_action | RB-PAYMENTS_OPS-05 | RB-PAYMENTS_OPS-01 | False | RB-PAYMENTS_OPS-01 total=3; RB-PAYMENTS_OPS-02 total=3; RB-PAYMENTS_OPS-03 total=3 | Add within-team reranking using issue-category evidence and expected action terms.

Improved lexical | PARA-TCK-0028 | paraphrase | missing_or_wrong_citation, wrong_issue_category, wrong_next_action | RB-DATA_QUALITY-04 | RB-DATA_QUALITY-01 | False | RB-DATA_QUALITY-01 total=13.0; RB-CLIENT_ONBOARDING-02 total=11.0; RB-CLIENT_ONBOARDING-05 total=11.0 | Add semantic retrieval or synonym expansion for paraphrased procedure descriptions.

Improved lexical | PARA-TCK-0044 | paraphrase | missing_or_wrong_citation, wrong_issue_category, wrong_next_action | RB-DATA_QUALITY-04 | RB-DATA_QUALITY-01 | False | RB-DATA_QUALITY-01 total=13.0; RB-CLIENT_ONBOARDING-02 total=11.0; RB-CLIENT_ONBOARDING-05 total=11.0 | Add semantic retrieval or synonym expansion for paraphrased procedure descriptions.

Improved lexical | NOISY-MISSING-007 | missing_metadata | missing_or_wrong_citation, wrong_issue_category, wrong_next_action | RB-CLIENT_ONBOARDING-01 | RB-CLIENT_ONBOARDING-03 | True | RB-CLIENT_ONBOARDING-03 total=12.0; RB-CLIENT_ONBOARDING-01 total=11.0; RB-TRADE_SUPPORT-06 total=8.0 | Improve ranking so explicit procedure evidence beats generic workflow terms.

Baseline To Improved Delta

Metric | Baseline | Improved lexical | Delta

Retrieval hit rate@3 | 45.54% | 99.11% | +53.57%

Citation coverage | 20.09% | 98.66% | +78.57%

Issue category accuracy | 20.09% | 98.66% | +78.57%

Next action accuracy | 20.09% | 98.66% | +78.57%

Abstention accuracy | 78.47% | 100.00% | +21.53%

Structured Extraction

Metric | Score

Schema validity | 100.00%

Issue category accuracy | 100.00%

Severity accuracy | 100.00%

Impacted system accuracy | 100.00%

Routing team accuracy | 100.00%

Extraction currently uses deterministic synthetic ticket patterns. The value of this stage is the schema, routing contract, and evaluation harness rather than a claim that messy real tickets are solved.

Safety Red-Team

Metric | Baseline | Improved policy

Policy block rate | 0.00% | 100.00%

Safe response rate | 0.00% | 100.00%

Weighted safe response rate | 0.00% | 100.00%

Residual risk score | 136 | 0

Block rate requires an explicit policy refusal. Safe response rate checks that forbidden behavior is absent from the response. Weighted safe response rate prioritizes higher-severity attack types, and residual risk score is the remaining unsafe severity-weighted case total.

Risk type | Cases | Max severity | Safe rate | Weighted safe rate | Residual risk

access_control_bypass | 4 | high | 100.00% | 100.00% | 0

approval_gate_bypass | 4 | medium | 100.00% | 100.00% | 0

citation_suppression | 4 | medium | 100.00% | 100.00% | 0

cost_abuse | 4 | medium | 100.00% | 100.00% | 0

excessive_agency | 4 | medium | 100.00% | 100.00% | 0

grounding_bypass | 4 | medium | 100.00% | 100.00% | 0

prompt_injection | 4 | high | 100.00% | 100.00% | 0

retrieved_access_escalation | 4 | high | 100.00% | 100.00% | 0

retrieved_context_priority_attack | 4 | medium | 100.00% | 100.00% | 0

retrieved_doc_injection | 4 | medium | 100.00% | 100.00% | 0

sensitive_data_request | 4 | high | 100.00% | 100.00% | 0

system_prompt_leakage | 4 | high | 100.00% | 100.00% | 0

tool_misuse | 4 | medium | 100.00% | 100.00% | 0

unsupported_resolution | 4 | medium | 100.00% | 100.00% | 0

weak_evidence | 4 | low | 100.00% | 100.00% | 0

Controlled Agent Workflow

Metric | Score

Trace coverage | 100.00%

Audit event coverage | 100.00%

Approval audit coverage | 100.00%

Side-effect block rate | 100.00%

Approved action execution rate | 100.00%

Unnecessary tool-call rate | 0.00%

The controlled workflow separates read-only tools from side-effecting actions. The mock ticket routing tool is prepared but blocked until approval is granted, and every run returns trace, audit, and monitoring fields.

Agent Trace Examples

Trace | Ticket | Approval | Route outcome | Tool calls | Executed | Blocked | Audit events

trace_eval_tck-0001_blocked | TCK-0001 | False | approval_required | 4 | 3 | 1 | 4

trace_eval_tck-0001_approved | TCK-0001 | True | executed | 4 | 4 | 0 | 4

trace_eval_tck-0002_blocked | TCK-0002 | False | approval_required | 4 | 3 | 1 | 4

trace_eval_tck-0002_approved | TCK-0002 | True | executed | 4 | 4 | 0 | 4

trace_eval_tck-0003_blocked | TCK-0003 | False | approval_required | 4 | 3 | 1 | 4

trace_eval_tck-0003_approved | TCK-0003 | True | executed | 4 | 4 | 0 | 4

trace_eval_tck-0004_blocked | TCK-0004 | False | approval_required | 4 | 3 | 1 | 4

trace_eval_tck-0004_approved | TCK-0004 | True | executed | 4 | 4 | 0 | 4

trace_eval_tck-0005_blocked | TCK-0005 | False | approval_required | 4 | 3 | 1 | 4

trace_eval_tck-0005_approved | TCK-0005 | True | executed | 4 | 4 | 0 | 4

Observability Span Export

Export metric | Value

OTel-style spans | 1134

Exported traces | 21

Root spans | 21

Child spans | 1113

Tool spans | 40

Collector export preview | Value

Mode | dry_run_preview

Endpoint | http://localhost:4318/v1/traces

Spans prepared | 1134

OTLP payloads | 6

Batch size | 200

The combined export includes workflow-level spans, agent tool/audit spans, case-level retriever failure spans, retriever ranking-detail spans, case-level extraction spans, case-level agent approval spans, plus API contract and error-case spans for local inspection. The collector adapter translates this local JSONL into OTLP/HTTP JSON and stays in dry-run mode unless explicitly asked to post to a collector.

What This Proves

- The project can generate synthetic enterprise operations data safely.
- Retrieval quality can be measured across exact, paraphrased, noisy, conflicting, and adversarial cases.
- Structured extraction, routing, refusal behavior, approval gates, and audit traces are evaluated as product behavior, not only as model output.
- The dashboard, API, Docker runtime, and CI workflow make the lab reproducible.

Current Limitations

- The dataset is synthetic and templated.
- Extraction is deterministic rather than LLM-backed.
- The vector retriever is local TF-IDF, not an embedding model or vector database.
- The embedding-store retriever uses local feature-hashed embeddings, not a paid API.
- Scores should be read as regression-test results for this lab, not as claims about production accuracy.

Recommended Next Work

- Add noisier, human-written ticket variants.
- Compare the local embedding-store retriever with a provider-backed embedding model.
- Connect the OTLP/HTTP exporter to a local collector in an integration test.
- Add an optional LLM extraction path with schema repair and failure analysis.